THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL UNIT TEST SET 1

Chapters 1–4: **The Wonderful World of Science, Diversity in the Living World, Mindful Eating: A Path to a Healthy Body, Exploring Magnets** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A student heats two beakers: one with 100 mL and one with 200 mL of water and notes the time to boil. Which statement about fairness is true? 1

(a) the volume of water is changed along with the time, so the test has two changing things
(b) the beakers must differ in colour
(c) time is the cause
(d) it is a perfectly fair test

2. A scientist finds a plant with parallel veins, fibrous roots and a soft stem. Which is most probable? 1

(a) a grass **(b)** a peepal tree **(c)** a mango tree **(d)** a rose bush

3. A food gives a blue-black colour with iodine and an oily patch on paper. It contains: 1

(a) protein and fibre **(b)** starch only
(c) fat only **(d)** starch and fat

4. Three identical-looking bars P, Q, R: P attracts Q at both ends, Q attracts R at both ends, R repels P at one end. Which can be concluded? 1

(a) P and R are magnets, Q is magnetic material
(b) all are plastic
(c) all are iron
(d) all are magnets

5. A study found that more ice cream is sold when more people get sunburnt. The best explanation is: 1

(a) both depend on hot weather **(b)** sunburn causes ice cream sales
(c) ice cream causes sunburn **(d)** it is magic

6. Which of the following is a correct inference about a tiger's eyes in front of the face? 1

(a) helps see all sides **(b)** helps fly
(c) helps judge distance for hunting **(d)** helps to swim

7. Why do nutritionists advise limiting packaged drinks? 1

(a) they have fibre
(b) they contain much sugar and few nutrients
(c) they contain protein
(d) they have vitamins

8. Which of the following is the best argument for using a magnet to separate iron from copper turnings? 1

(a) iron is magnetic and copper is not **(b)** both are shiny
(c) both are metals **(d)** both rust

9. Suppose two groups test the same hypothesis and get opposite results. What should be done first? 1

(a) believe one group **(b)** toss a coin
(c) ignore both **(d)** compare the methods and repeat carefully

10. In a forest, if all insects vanish, what is most likely to happen? 1

(a) forest will grow faster **(b)** nothing will change
(c) many plants will fail to pollinate **(d)** animals will grow bigger

11. The test: a food sample is crushed, mixed with water, and copper sulphate and caustic soda are added. A violet colour appears. This shows: 1

(a) starch **(b)** protein **(c)** fibre **(d)** fat

12. An iron nail is rubbed with the north pole of a magnet in one direction many times. The end where the stroke ends becomes a: 1

(a) no pole **(b)** south pole **(c)** north pole **(d)** neutral

13. In an experiment on how salt affects the boiling point, two things were changed: salt amount and heater power. The error is: 1

(a) salt is a solid **(b)** water is cold
(c) too many readings **(d)** more than one variable was changed

14. If an animal has webbed feet, gills in young stage and lungs in adult stage, it is a(n): 1

(a) mammal **(b)** amphibian **(c)** bird **(d)** reptile

15. A student tested three foods P, Q, R with iodine. Only Q turned blue-black. Which statement is correct? 1

(a) Q contains starch **(b)** All contain starch
(c) P and R contain starch **(d)** None contain anything

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. In an experiment on plant growth, A is kept in sunlight and B in shade, but A is also given more water. Why is this unfair? 2

17. Cactus leaves are spines but it still makes food. How? 2

18. A food sample turns blue-black with iodine and leaves an oily spot on paper. What conclusions can you draw? 2

19. A compass is placed on a steel table and gives a wrong reading. Explain and suggest a remedy. 2

20. How can a scientific model such as a globe help us? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Design a fair test to compare which of three fabrics dries the fastest. Name controlled variables. 3

22. A plant has leaves with net veins, a tap root, a thick woody trunk 15 m tall. Classify it and give the chief features of each part. 3

23. A girl's diet: breakfast - bread and jam, lunch - noodles, dinner - fried snacks. Evaluate it and suggest changes. 3

24. Describe how to show that the poles of a magnet are not at the very ends but near them, using iron filings. 3

OR

Explain the role of curiosity, observation and reasoning in the discovery of gravity by Newton. 3

25. A student suggests classifying animals by colour. Why is this not a good method? 3

OR

Why should dairy foods be included in a school child's diet? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A student wants to know if the colour of a cup affects how fast hot water cools. Design the investigation. 5

OR

Design a chart classifying 10 organisms by
(i) habitat and
(ii) food habit, and explain how features match. 5

27. A group of students are tested for health. Student A has bleeding gums, B has night blindness, C has swollen neck, D has bent legs, E is pale. Identify the cause, the remedy for each. 5

OR

Design an investigation to compare the strength of two magnets using paper clips and describe how to make it a fair test. 5

END OF QUESTION PAPER — Total: 50 Marks

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